

In [1]: *# step 1 import libraries*

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

In [3]: *# step 2 load dataset*

```
data = pd.read_csv("data/house_data.csv")
data.columns = data.columns.str.strip()
data.head()
```

Out[3]:

	House Size	Bedrooms	Bathrooms	Age	Distance from City	Price
--	------------	----------	-----------	-----	--------------------	-------

0	500	1	1	10	12	25
1	800	2	1	8	10	40
2	1000	2	2	6	8	50
3	1200	3	2	4	5	65
4	1500	3	3	2	3	90

In [5]: *# step 3 select input columns and output column*

```
X = data[['House Size',
          'Bedrooms',
          'Bathrooms',
          'Age',
          'Distance from City']]

y = data['Price']
```

In [6]: *# step 4 split into training and testing sets*

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.4, random_state=42
)
```

In [7]: *# step 5 train linear regression model*

```
model = LinearRegression()
model.fit(X_train, y_train)
```

Out[7]:

▼ LinearRegression ⓘ ?

► Parameters

In [8]: *# step 6 get coefficients and intercept*

```
intercept = model.intercept_  
coefficients = pd.DataFrame({  
    "Feature": X.columns,  
    "Coefficient": model.coef_  
})  
  
intercept, coefficients
```

Out[8]: (np.float64(53.03212708416973),
 Feature Coefficient
 0 House Size 0.021969
 1 Bedrooms 1.136334
 2 Bathrooms -0.757483
 3 Age -0.757702
 4 Distance from City -2.651518)

In [9]: *# step 7 predict test results*

```
y_pred = model.predict(X_test)
```

In [10]: *# step 8 compare actual and predicted prices*

```
results = pd.DataFrame({  
    "Actual Price": y_test.values,  
    "Predicted Price": y_pred  
})  
  
results
```

Out[10]:

	Actual Price	Predicted Price
0	40	39.545335
1	90	77.651518

In [11]: *# step 9 evaluate model performance*

```
mse = mean_squared_error(y_test, y_pred)  
r2 = r2_score(y_test, y_pred)  
  
mse, r2
```

Out[11]: (76.34585818363335, 0.8778466269061866)

In [15]: *# step 10 visualize improved actual vs predicted*

```
plt.figure(figsize=(8,6))  
  
plt.scatter(y_test, y_pred)  
  
min_val = min(min(y_test), min(y_pred))  
max_val = max(max(y_test), max(y_pred))
```

```
plt.plot([min_val, max_val],
         [min_val, max_val],
         linestyle='--')

plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted House Prices")

plt.grid(True, linestyle=':', linewidth=0.7)

plt.xlim(min_val - 5, max_val + 5)
plt.ylim(min_val - 5, max_val + 5)

plt.savefig("linear_regression_results.png", dpi=300, bbox_inches="tight")
plt.show()
```

